



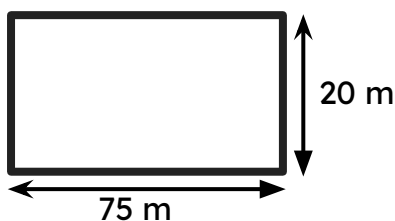
Agricultural worker (animals)

Task A: Geometry in agricultural work

1) For each of the following calculate the maximum number of each animal which can be safely and comfortably looked after in these barns:

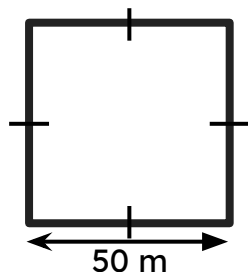
Diagrams are
not drawn to
scale

a)

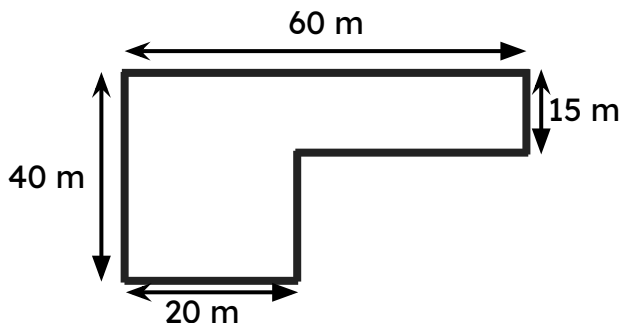


	Animal	Required living area
i	Cows	10 m ²
ii	Chickens	1 m ²
iii	Pigs	2.2 m ²
iv	Sheep	2 m ²

b)



c)



Task B: Costs and percentages

1) A farmer decides that they want to give their cows more space.

Cost of care for a cow per day is £4

They are going to give each cow 12 m^2 .

Assume each cow produces 20 litres of milk a day.

The selling price per litre of milk is 38 p.

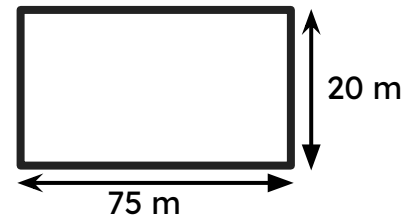
Calculate the following:

a) The number of cows the barn can house.

b) The cost of care for the cows per year.

c) The income from milk production.

d) The percentage profit.



2) A farmer decides that they want keep chickens.

They are going to give each chicken an area of 15 m^2

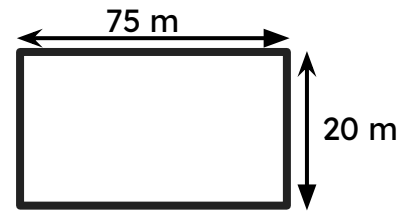
Cost of care for a chicken per year is £50

Each chicken produces 0.8 eggs on average per day.

They decide to sell the eggs themselves, selling 1 dozen for £3.20

Calculate the following:

a) The percentage profit.



b) The farmer wants make a 60% profit. How much do they need to charge for a dozen eggs?